

Exercise Sheet 1: Sample Spaces, σ -Algebras and Probability

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About this sheet. There are seventeen questions in this exercise sheet. Write the sample space and the events clearly before computing probabilities. Use descriptive language when writing your answers. Do *not* rely only on symbols to convey your meaning.

Source. The sheet is based mainly on Chapter 1 of Rohatgi and Saleh, with the wording simplified and some questions modified or added.

Part A. Sample spaces and σ -algebras

1. Write down the sample space Ω explicitly for each experiment.

- (a) A coin is tossed three times.
- (b) A die is rolled until a 6 appears for the first time.
- (c) Two cards are drawn one after another from a deck of 52 cards, without replacement. The order is recorded.
- (d) Repeat the previous exercise when the order is *not* recorded.
- (e) The lifetime, in hours, of a light bulb is recorded.

For each part, say whether Ω is finite, countably infinite, or uncountable.

2. A coin is tossed three times.

- (a) Write down Ω .
- (b) Write down the event A : exactly one head occurs.
- (c) Write down the event B : at least one head occurs.
- (d) Write down the event C : the first toss is a head and the third toss is a tail.

3. Let $\mathcal{S}$ be a σ -algebra on Ω . Prove that if $A, B \in \mathcal{S}$, then $A - B \in \mathcal{S}$.

4. Let $\mathcal{S}_1$ and $\mathcal{S}_2$ be σ -algebras on Ω . Prove that $\mathcal{S}_1 \cap \mathcal{S}_2$ is a σ -algebra on Ω .

5. Extend the previous exercise to an arbitrary collection of σ -algebras. That is, suppose that I is an arbitrary indexing set, and $\{\mathcal{S}_i : i \in I\}$ is a collection of σ -algebras on Ω . Show that $\bigcap_{i \in I} \mathcal{S}_i$ is a σ -algebra on Ω .

6. Let $\Omega = \{1, 2, 3, 4\}$. Decide whether each collection below is a σ -algebra. Give a reason.

$$\mathcal{S}_1 = \{\emptyset, \{1, 2\}, \{3, 4\}, \Omega\},$$

$$\mathcal{S}_2 = \{\emptyset, \{1\}, \{2, 3, 4\}, \{1, 2\}, \Omega\}.$$

If a collection is not a σ -algebra, state exactly which property fails.

7. Let

$$\Omega = \{1, 2, 3, 4, 5, 6\}, \quad B_1 = \{1, 2\}, \quad B_2 = \{3\}, \quad B_3 = \{4, 5, 6\}.$$

List all sets in the smallest σ -algebra containing B_1, B_2, B_3 .

8. Let Ω be a set, let $(\Omega', \mathcal{S}')$ be a sample space, and let

$$f : \Omega \rightarrow \Omega'$$

be a function. Define

$$\mathcal{S} = \{f^{-1}(A) : A \in \mathcal{S}'\}.$$

Show that $\mathcal{S}$ is a σ -algebra on Ω .

Part B. Probability spaces and computations

9. Let $(\Omega, \mathcal{S}, P)$ be a probability space. Starting from the probability axioms, prove:

- (a) if $A \subseteq B$, then $P(A) \leq P(B)$;
- (b) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- (c) $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$.

10. A box contains 1000 light bulbs. Suppose

$$P(\text{at least one defective bulb}) = 0.10, \quad P(\text{at least two defective bulbs}) = 0.05.$$

Find the probability that the box contains:

- (a) no defective bulb;
- (b) exactly one defective bulb;
- (c) at most one defective bulb.

11. Let

$$\Omega = \{1, 2, 3, 4, 5, 6\}, \quad \mathcal{S} = \mathcal{P}(\Omega).$$

Define

$$P(\{k\}) = \frac{k}{21}, \quad k = 1, \dots, 6,$$

and, for $A \subseteq \Omega$, define

$$P(A) = \sum_{k \in A} P(\{k\}).$$

- (a) Verify that $(\Omega, \mathcal{S}, P)$ is a probability space.
- (b) Find $P(\{2, 4, 6\})$.
- (c) Find $P(\{k : k \geq 4\})$.

12. Let

$$\Omega = \{1, 2, 3, 4\}, \quad \mathcal{S} = \{\emptyset, \{1, 2\}, \{3, 4\}, \Omega\}.$$

For $0 \leq p \leq 1$, define

$$P(\{1, 2\}) = p, \quad P(\{3, 4\}) = 1 - p,$$

with $P(\emptyset) = 0$ and $P(\Omega) = 1$.

- (a) Verify that $(\Omega, \mathcal{S}, P)$ is a probability space.
- (b) Why is it not necessary to assign a value to $P(\{1\})$?

13. Let

$$\Omega = \{1, 2, 3, \dots\}, \quad \mathcal{S} = \mathcal{P}(\Omega), \quad P(\{n\}) = \frac{1}{2^n}.$$

- (a) Verify that the probabilities of all one-point events add up to 1.
- (b) Find the probability that n is even.
- (c) Find the probability that n is a multiple of 3.
- (d) Find $P(\{n : n \geq 5\})$.

14. Two fair dice are rolled. Take

$$\Omega = \{(i, j) : 1 \leq i, j \leq 6\},$$

with all 36 outcomes equally likely. Let

$$A = \{(i, j) : i + j \geq 10\}, \quad B = \{(i, j) : i = j\}.$$

Find

$$P(A), \quad P(B), \quad P(A \cap B), \quad P(A \cup B).$$

15. A point (X, Y) is chosen uniformly from the unit square $[0, 1] \times [0, 1]$. Probability is given by area. Find the probability that:

- (a) $X + Y \leq 1$;
- (b) $X \leq \frac{1}{2}$ and $Y \geq \frac{1}{2}$;
- (c) $|X - Y| \leq \frac{1}{2}$.

Draw the required regions before calculating their areas.

16. A point is chosen uniformly from $[0, 1]$.

- (a) Find $P(\{1/2\})$.
- (b) Show that every one-point set $\{x\}$ has probability 0.
- (c) Using countable additivity, show that

$$P(\mathbb{Q} \cap [0, 1]) = 0.$$

- (d) Hence find the probability that the chosen point is irrational.

Does an event of probability 0 have to be the empty set?

Part C. Independence of events

Definition. Two events A and B are called *independent* if

$$P(A \cap B) = P(A)P(B).$$

Use this definition in Question 14. No conditional probability is needed.

17. (a) Two fair coins are tossed. Let

$$A = \{\text{first toss is a head}\}, \quad B = \{\text{second toss is a head}\}.$$

Are A and B independent?

(b) One card is chosen from a standard deck of 52 cards. Let

$$A = \{\text{the card is an ace}\}, \quad B = \{\text{the card is a club}\}.$$

Are A and B independent?

(c) Suppose A and B are disjoint and $P(A) > 0$, $P(B) > 0$. Can A and B be independent? Explain.

How to present your work. Use set notation clearly. When you claim that a collection is a σ -algebra or that a function is a probability, check the relevant axioms. For geometric probability questions, draw the sample space and the event before calculating an area.